

MENTORA ACADEMY

EXCELLENCE IN EDUCATION

Edexcel International A Level Chemistry (WCH14)

UNIT 4 — TOPICAL WORKSHEET PACK: 15D.2

CARBOXYLIC ACID DERIVATIVES: ESTERS

CANDIDATE NAME:	Usman
SUBJECT & CODE:	Chemistry Unit 4 — Rates, Equilibria & Organic (WCH14)
TOPIC AREA:	Topic 15: CARBOXYLIC ACID DERIVATIVES, SPECTROSCOPY AND CHROMATOGRAPHY
SUB-TOPIC:	15D.2: Carboxylic Acid Derivatives: Esters
TOTAL QUESTIONS:	2 Questions (Structured + MCQs + Calculations)
TOTAL MARKS:	5 Marks
EXAMINER FAQs:	1 Empirical Examiner FAQs & Traps Included

HOW TO USE THIS TOPICAL PACK:

1. Complete all structured questions under timed exam conditions.
2. Pay close attention to command words (Explain, Deduce, Calculate, State).
3. Review the complete worked mark scheme in Section C.
4. Study the Empirical Examiner FAQs in Section D to avoid common candidate traps.

SECTION A: TOPICAL PRACTICE QUESTIONS — 15D.2

Q1. 1. (a) Write the equation for the acid-catalysed hydrolysis of ethyl ethanoate with dilute H_2SO_4 .

[2 Marks]

WCH14/01/Oct22/Q7(a)

Q2. (b) Contrast the acid-catalysed hydrolysis with alkaline hydrolysis (saponification) using aqueous NaOH .

[3 Marks]

WCH14/01/Oct22/Q7(b)

SECTION B: WORKED MARK SCHEME — 15D.2

Q	MARKING SCHEME & ACCEPTABLE ANSWERS	MARKS
Q1	Q1: 1. (a) Write the equation for the acid-catalysed hydrolysis of ethyl ethanoate with dilute H ₂ SO ₄ . 1. $\text{CH}_3\text{COOCH}_2\text{CH}_3 + \text{H}_2\text{O} \rightarrow \text{CH}_3\text{COOH} + \text{CH}_3\text{CH}_2\text{OH}$. 2. Reversible equilibrium reaction.	[2]
Q2	Q2: (b) Contrast the acid-catalysed hydrolysis with alkaline hydrolysis (saponification) using aqueous NaOH. 1. $\text{CH}_3\text{COOCH}_2\text{CH}_3 + \text{NaOH} \rightarrow \text{CH}_3\text{COONa} + \text{CH}_3\text{CH}_2\text{OH}$. 2. Alkaline hydrolysis goes to completion (irreversible) yielding sodium carboxylate salt. 3. Gives higher yield of alcohol product than acid hydrolysis.	[3]

SECTION C: EMPIRICAL EXAMINER FAQs & COMMON TRAPS — 15D.2**FAQ #1: Why is alkaline hydrolysis of esters preferred over acid hydrolysis for preparation?****Category: Reaction Yields****COMMON EXAMINER TRAP / MISTAKE:**

Stating acid hydrolysis gives a higher yield.

MODEL ANSWER / EXAMINER EXPECTATION:

Acid hydrolysis is reversible and reaches an equilibrium position (lower yield). Alkaline hydrolysis is irreversible because the carboxylate ion (CH_3COO^-) is unreactive towards nucleophilic attack by the alcohol, driving the reaction 100% to completion.